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Optimizing CPR performance: the role of body composition and ergonomic positioning

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Abstract

Background Quality cardiopulmonary resuscitation (CPR) is associated with improved survival rates in prehospital arrest cases. Although CPR guidelines emphasize compression depth, rate, and recoil, individual anatomical and biomechanical factors influencing CPR performance remain underexplored. This study aims to determine the optimal CPR position by focusing on individual positional conditions and anthropometric data to improve CPR effectiveness and guide team member selection in emergency scenarios.

Methods A cross-sectional study involving 123 paramedic associate degree students was conducted. Anthropometric measurements including upper extremity length, biceps circumference, and femur length were obtained. CPR performance was assessed using a feedback-capable mannequin, with data collected on compression depth, frequency, rhythm quality, and compression/decompression ratios. Positional parameters such as knee and hip angles, distance between knees, and distance from the mannequin were recorded and analyzed in relation to CPR outcomes.

Results Male participants demonstrated significantly higher CPR quality scores than females (65.09 ± 28.26 vs. 48.85 ± 27.11, $p = 0.004$). Positive correlations were found between BMI ($r = 0.239$, $p = 0.008$), upper extremity length ($r = 0.364$, $p < 0.001$), biceps circumference ($r = 0.350$, $p < 0.001$), and CPR quality metrics. A moderate inter-knee position (shoulder-width distance) provided superior compression depth and rhythm stability compared to narrow or wide positions ($p < 0.001$).

Conclusion Anthropometric and positional factors significantly influence CPR performance. Emergency team members with appropriate physical profiles (e.g., greater muscle mass, limb length, BMI) and adopting occupational ergonomic factors (e.g., moderate knee range, limited joint flexion) may improve CPR quality in the prehospital setting.

Keywords Biomechanics, Cardiopulmonary resuscitation, Ergonomics, Heart massage

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Introduction

Cardiopulmonary resuscitation (CPR) is an essential emergency procedure to sustain blood circulation and preserve brain function in people who suffer from cardiac arrest. Until spontaneous cardiac function can be restored or more advanced medical care becomes available, this life-saving method uses rescue breathing and chest compressions to assist oxygenation and circulation [1]. In order to improve the likelihood of survival and lower the danger of irreversible neurological injury, the fundamental goal of CPR is to maintain a partial flow of oxygenated blood to the brain and essential organs [2].

CPR is still one of the most popular and successful first-aid treatments for out-of-hospital cardiac arrest, even with major improvements in emergency medicine and technology. According to studies, performing prompt, effective CPR can increase a person's chances of survival by two or even three times [3]. Furthermore, continuous worldwide training programs emphasize the critical role that cardiopulmonary resuscitation plays in the chain of survival and seek to raise public awareness and preparedness for performing it [4].

The paramedic profession requires a high level of technical competence, the ability to make rapid decisions, and the capacity to act swiftly and accurately. Paramedics perform emergency life support procedures, such as CPR, in pre-hospital cardiac arrests [5]. High-quality chest compressions are associated with higher survival rates in patients experiencing cardiac arrest [6]. The quality of CPR can be assessed through parameters such as the number of compressions per minute, compression and decompression ratios, and compression depth [7]. Studies examine the quality of CPR from various perspectives; such as performing CPR while seated or standing [8], based on the height of the platform [9], according to the position of the dominant hand [10, 11], using technological devices that provide feedback [12] and conducting effective CPR for 2 min with 1-minute rotations [13]. These studies, from a biomechanical perspective, address the scene and the application to guide practitioners toward an ergonomic CPR technique that causes less fatigue over longer periods. This study aims to identify the most optimal CPR position based on angular assessments and anthropometric data, focusing on individual positioning. This will help in selecting the most suitable person from the team for CPR and guide in delivering effective compressions for longer durations in the most appropriate position.

Materials and methods

The study was conducted retrospectively in accordance with the World Medical Association Declaration of Helsinki Ethical Principles for Medical Research Involving Human Participants. Approval for the study was obtained from the Ethics Committee of the Health Sciences Department of Karadeniz Technical University (No: 2024/90, Date: 03.06.2024). Informed consent forms were obtained from all participants after providing the necessary information.

Participants

The study included 123 undergraduate paramedic students (88 females, 35 males) from Giresun University Şebinkarahisar Health Services Vocational School. The participants were comprised of 2nd year students who were more experienced and had frequent resuscitation practice, and 1st year students who were relatively inexperienced and had less practice. Inclusion criteria were active enrollment in a paramedic program, completion of basic CPR training, age 18–25 years, and absence of musculoskeletal disorders affecting performance. Exclusion criteria included previous CPR instructor certification, recent upper extremity injury, and inability to kneel for extended periods. Data on age, gender, weight, height, and class level were collected from all participants. BMI was calculated (kg/m^2). The CPR environment consisted of a CPR mannequin placed on the floor and a monitoring system.

CPR performance evaluation

Equipment: CPR procedures were performed using a Laerdal Resusci Anne® QCPR manikin equipped with Bluetooth connectivity and real-time feedback capability. The manikin was placed on the floor to simulate realistic prehospital conditions. This enabled us to collect quantitative data such as total CPR quality, compression depth, average compressions per minute, rhythm of CPR speed, and compression and decompression percentages (Table 1; Fig. 1).

Biomechanical data

Upper extremity length (from the glenoid prominence to the wrist joint), maximum biceps circumference (at the thickest point of the brachial region), and femur length (from the greater trochanter to the lateral epicondyle) were measured using a standard tape measure. During the CPR procedure, photographs and video recordings were taken from the front and side views at the same level as the manikin using a tripod. Camera positions were standardized across all participants. The position of the provider during CPR (hip and knee joint angles) was determined from these recordings. The distance from the knees to the model was recorded. The distance between the knees was assessed using images taken from behind the participant (Fig. 2).

Table 1 CPR performance evaluation criteria and descriptions

CPR quality measurements: The Laerdal QCPR system provided validated, objective measurements [14, 15]	
Total CPR Quality Score (0-100%)	The cumulative score is calculated using Laerdal's algorithm, which measures closeness to guideline recommendations
Average Compression Depth Quality (0-100%)	Percentage of compressions reaching a depth of 50–60 mm
Compressions Per Minute	Actual number of compressions
Rhythm Quality (0-100%)	Consistency with compression intervals
Compression Rate (0-100%)	Percentage of chest compression phase per cycle
Decompression Rate (0-100%)	Percentage of chest recoil phase per cycle
Performance Protocol	Each participant performed continuous chest compressions for 2 min without ventilation, following established protocols for QCPR assessment.

CPR, cardiopulmonary resuscitation

Statistical analysis

Based on previous CPR quality studies with effect sizes of 0.5, $\alpha = 0.05$, and power = 0.80, the minimum required sample size was calculated as 64 participants. Our sample of 123 provides adequate power (> 90%) for detecting meaningful correlations. The research data were evaluated using SPSS 27.0 (Inc., Chicago, IL). Descriptive statistics were presented as mean ± SD, frequency distribution, and percentage. The conformity of the variables to normal distribution was examined. For variables found to be normally distributed, Student's t-test was used between two independent groups, and ANOVA was used in more than two independent groups. Pearson's correlation was used in determining the relationship between different areas. Statistical significance level was accepted as $p < 0.05$.

Results

Participant characteristics

Table 2 presents demographic characteristics and CPR performance metrics by gender. Male participants were significantly taller (178.86 ± 6.42 vs. 163.51 ± 5.18 cm, $p < 0.001$), heavier (74.72 ± 14.21 vs. 60.09 ± 9.95 kg, $p < 0.001$), and had longer upper extremities (59.67 ± 3.34 vs. 53.35 ± 3.06 cm, $p < 0.001$) compared to females. No significant differences were observed in age or BMI between genders.

Correlation of CPR performance with anthropometric data

It was observed that BMI had a positive correlation with total CPR quality ($r = 0.239$, $p = 0.008$) and compression quality ($r = 0.271$, $p = 0.002$). Regarding the relationship between positional variables and CPR outcomes, a negative correlation was observed between hip joint angle and compression depth ($r = -0.186$, $p = 0.039$), between knee joint angle and total CPR quality ($r = -0.200$, $p = 0.027$), and between hip and knee joint angles ($r = -0.328$, $p < 0.001$).

The maximum circumference of the biceps region showed a positive correlation with compression quality ($r = 0.454$, $p < 0.001$), total CPR quality ($r = 0.350$, $p < 0.001$), and compression ratio ($r = 0.456$, $p < 0.001$), while showing a negative correlation with decompression ratio ($r = -0.214$, $p = 0.017$).

Upper extremity length was positively correlated with total CPR quality ($r = 0.364$, $p < 0.001$), average compression depth quality ($r = 0.617$, $p < 0.001$), and compression ratio ($r = 0.618$, $p < 0.001$) (Table 3).

When the positional conditions affecting the CPR quality were examined, no statistically significant results were found regarding the proximity to the model. However, when the distance between the knees was evaluated in three groups (narrow, mean, and wide), the statistically significant relationships with CPR outcomes are presented in Table 4.

Discussion

This study focused on anatomical, ergonomic, and biomechanical reasons for CPR quality and showed that both anthropometric characteristics and positioning strategies significantly affect CPR quality. Our findings suggest that providers with longer upper limbs, greater muscle mass, and moderate BMI demonstrate better CPR quality. Certain positioning strategies, including moderate knee-to-knee distance and reduced joint flexion angles, increase compression effectiveness regardless of individual characteristics. All these data were compared with outcomes that assessed CPR quality from different perspectives and their relationships were examined.

Both the American Heart Association (AHA) and the European Resuscitation Guidelines (ERC) emphasize the importance of high-quality CPR during resuscitation. High-quality CPR is known to increase survival rates after cardiac arrest [16, 17]. According to current AHA guidelines for CPR administration, a person must compress the chest at least 2 inches (approximately 5 cm) (1/3 of the anterior-posterior thoracic cross-section of the body) at a rate of at least 100 chest compressions per minute (100–120/min) to be considered effective in creating blood flow in the patient [18]. Anderson et al. defines perfect CPR as a two-minute CPR session with a depth of 50–60 millimeters, a rate of 100–120 per minute, and complete chest retraction, with 90% or more compression [19]. In our study and the model we worked on, data were obtained by calibrating these standards.

Another issue affecting the quality of CPR is the position of the practitioner and which hand is placed at the bottom for compression according to that position.

Fig. 1 CPR outputs of the Laerdal Resusci Anne® QCPR brand resuscitation dummy. All data other than ventilation was monitored instantly on the screen. Ventilation was not applied during the application

According to the results, practices where the right hand is placed at the bottom in the left placement and practices where the left hand is placed at the bottom in the right placement significantly affect the CPR compression pressure and depth [20].

The strong correlation between upper limb length and multiple CPR measurements demonstrates the mechanical advantage provided by longer lever arms. Providers with longer arms can reach target compression depths (50–60 mm) with less effort and maintain CPR quality for longer periods of time. This result may be effective in decision-making for team or provider selection in prolonged CPR scenarios.

Although the kinematics of CPR varies significantly with the changing position of the provider, it has been shown that these differences do not affect the compression force, depth, and frequency applied by experienced providers. Therefore, whether the patient is on the floor or beds of different heights [21], it has been reported that this does not affect the quality of CPR [22]. However,

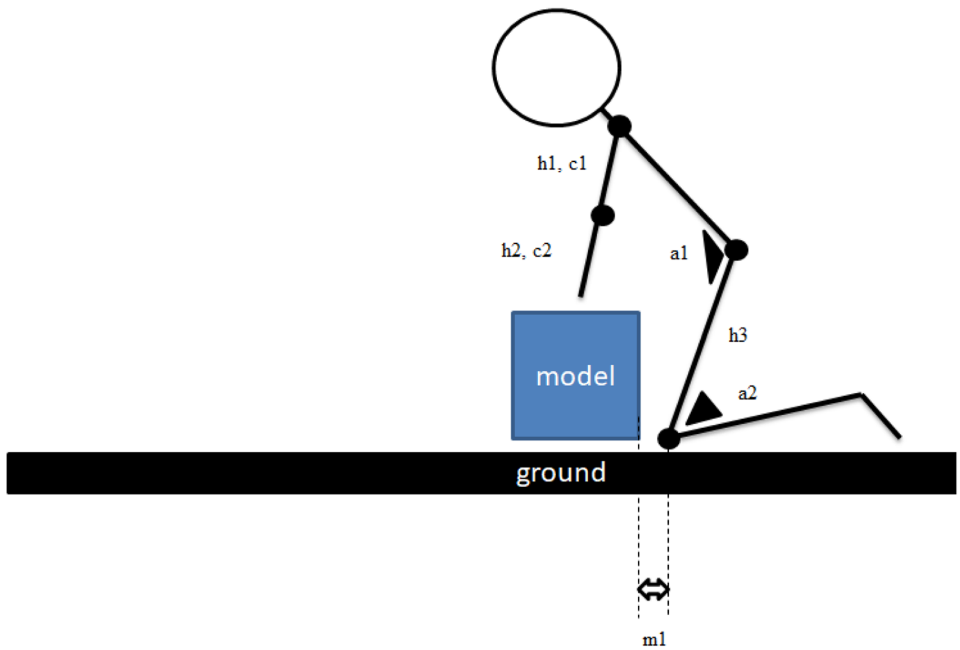

Fig. 2 Anatomical data obtained from the lateral view illustration of the CPR provider is shown. **a1**: hip joint angle, **a2**: knee joint angle, sum of **h1 and h2**: upper extremity length, **h3**: femur length, **c1 and c2**: local maximum circumference measurements, **m1**: distance to the model

Table 2 Demographic characteristics and comparison of variables between sexes

	Female ($n=88$)	Male ($n=35$)	*P*-value
Demographics			
Class 1	61	25	0.818
Class 2	27	10	
Age	20.61 ± 3.12	20.57 ± 0.91	0.938
Weight	60.09 ± 9.95	74.72 ± 14.21	< 0.001
Height	163.51 ± 5.18	178.86 ± 6.42	< 0.001
BMI	22.46 ± 3.45	23.40 ± 4.68	0.223
Upper extremity length	53.35 ± 3.06	59.67 ± 3.34	< 0.001
CPR performance metrics			
CPR total quality	48.85 ± 27.11	65.09 ± 28.26	0.004
Average compression depth quality	42.50 ± 7.18	55.54 ± 9.64	< 0.001
Compressions per minute	124.25 ± 11.99	128.69 ± 11.40	0.063*
Rhythm quality	32.76 ± 26.97	31.14 ± 33.54	0.780
Compression ratio	21.07 ± 23.73	65.29 ± 29.91	< 0.001
Decompression ratio	85.49 ± 19.15	59.26 ± 27.02	< 0.001

BMI, body mass index; CPR, cardiopulmonary resuscitation

* indicates statistically significant

another study reported that the position in which the biomechanical stress of the back muscles and lumbar region is minimized is in CPR performed on 60 cm high Table [23]. Nicolau et al., who evaluated the relationship between the model-provider limb angle and the quality of CPR applied in different positions during application, reported once again that the 90-degree angulation is the most efficient position [24]. In a study conducted by Hyun et al., a 20.3 cm distance between the knees gave more positive CPR quality results than a 50 cm distance [25].

Our finding that smaller hip and knee angles improve CPR performance aligns with biomechanical principles of force transmission. Hip angles < 70° and knee angles < 90° appear to optimize the kinetic chain from torso to hands, allowing more efficient transfer of body weight into compression force. This finding is supported by recent research on rescuer positioning that emphasizes the importance of ergonomic considerations in CPR delivery. According to Dainty, in continuous CPR applications (CCR CPR, continuous cardiac massage), significant decreases in CPR force occur from the beginning [26]. Based on the results, we can recommend that practitioners should perform arm strength exercises to improve CPR quality. This is in line with the data presented by López-González et al. [27]. In addition, the positive correlation of BMI with CPR quality is another parameter mentioned by Russo et al. [28].

Conclusion

Muscle volume, limb length and BMI play an important role in the selection of the CPR practitioner within the paramedic team. In addition, when the practitioners apply heart massage in the kneeling position on the floor, they position their knees at a moderate level, that is, at shoulder width, and do not allow high extension in the hip and knee joint angles, which has shown positive results in terms of CPR quality and compression depth.

Table 3 Correlation of CPR quality parameters with anthropometric data

Anthropometric variable	Total CPR quality	Compression depth quality	Compression per minute	Rhythm quality	Compression ratio	Decom-pres-sion ratio
Height	0.256** ($p=0.004$)	0.593*** ($p<0.001$)	0.191* ($p=0.034$)	-0.012 ($p=0.897$)	0.641*** ($p<0.001$)	-0.449*** ($p<0.001$)
Weight	0.341*** ($p<0.001$)	0.545*** ($p<0.001$)	-0.022 ($p=0.806$)	0.122 ($p=0.179$)	0.562*** ($p<0.001$)	-0.410*** ($p<0.001$)
BMI (kg/m^2)	0.239** ($p=0.008$)	0.271** ($p=0.002$)	-0.132 ($p=0.146$)	0.130 ($p=0.153$)	0.267** ($p=0.003$)	-0.202* ($p=0.025$)
Upper extremity Length (cm)	0.364*** ($p<0.001$)	0.617*** ($p<0.001$)	0.017 ($p=0.850$)	0.101 ($p=0.265$)	0.618** ($p<0.001$)	-0.340** ($p<0.001$)
Biceps Circumference (cm)	0.350*** ($p<0.001$)	0.454*** ($p<0.001$)	-0.050 ($p=0.580$)	0.080 ($p=0.381$)	0.4567*** ($p<0.001$)	-0.214* ($p=0.017$)

BMI, body mass index; CPR, cardiopulmonary resuscitation

* $p<0.05$, ** $p<0.01$, *** $p<0.001$

Table 4 Effect of the distance between the two knees positioned during CPR application on CPR outputs

	Narrow (*n*=81)	Mean (*n*=26)	Wide (*n*=16)	*P*-value	Post-hoc significance
CPR total quality	50.91 ± 28.88	63.58 ± 26.70*	50.00 ± 25.59	< 0.001	moderate > narrow, wide
Average compression depth quality	43.88 ± 8.90	53.92 ± 8.83*	45.50 ± 10.40	< 0.001	moderate > narrow, wide
Compressions per minute	123.7 ± 12.06	125.88 ± 10.09	134.0 ± 11.08*	0.006	wide > moderate, narrow
Rhythm quality	33.70 ± 26.77	40.15 ± 33.98*	12.44 ± 22.15	0.007	moderate > wide
Compression ratio	26.96 ± 29.35	53.77 ± 30.17*	34.81 ± 38.79	< 0.001	moderate > narrow, wide
Decompression ratio	81.52 ± 22.99	65.31 ± 28.93*	81.00 ± 19.08	0.011	narrow, wide > moderate

CPR, cardiopulmonary resuscitation

* indicates statistically significant

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Author contributions
TA: Conceptualization, Methodology, Visualization, Formal Analysis, Writing – Original Draft; OK: Data curation, Resources; HS: Visualization, Resources, Data curation, Investigation; AK: Formal Analysis, Supervision; NY: Methodology, Visualization, Formal Analysis; GH: Supervision, Writing- Reviewing and Editing.

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Data availability
All data are available upon request from the first author, Dr. Tayfun Aygun, by contacting fzttayfunaygun@gmail.com.

Declarations

Ethical approval
The study was conducted retrospectively in accordance with the World Medical Association Declaration of Helsinki Ethical Principles for Medical Research Involving Human Participants. Approval for the study was obtained from the Ethics Committee of the Health Sciences Department of Karadeniz Technical University (No: 2024/90, Date: 03.06.2024).

Consent to participate
Informed consent forms were obtained from all participants after providing the necessary information.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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